

**EFFECT OF INDIUM SUBSTITUTION
ON THE STRUCTURE OF $\text{SrFeO}_{3-\delta}$ AND ITS TRANSPORT PROPERTIES**

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Materials based on $\text{SrFeO}_{3-\delta}$ are promising candidates for cathodes in solid oxide fuel cells and oxygen-permeable membranes. However, undoped $\text{SrFeO}_{3-\delta}$ exhibits a high thermal expansion coefficient and is prone to phase transformations in a reducing atmosphere, which limits its practical application. The introduction of dopants into the B-sublattice enables control over structural stability, oxygen nonstoichiometry, and transport properties. This approach is realized in the present work by substituting $\text{Fe}^{3+/4+}$ with In^{3+} and studying its effect on these characteristics.

Samples with the formula $\text{SrFe}_{1-x}\text{In}_x\text{O}_{3-\delta}$ ($x = 0.00\text{--}0.35$) were synthesized using a sol-gel method with citric acid, followed by annealing at 1100 °C and sintering at 1190 °C. The phase composition of the samples was studied by X-ray diffraction with Rietveld refinement. The homogeneity range is within $0.0 \leq x \leq 0.30$. The samples with an indium content of $x = 0.05\text{--}0.30$ crystallize in a cubic perovskite cell (space group $Pm\text{-}3m$), whereas the base $\text{SrFeO}_{3-\delta}$ composition possesses an orthorhombic structure (space group $Cmmm$). Cubic symmetry can exist for the undoped composition, however, this modification is unstable under standard conditions. Thus, the introduction of indium stabilized the cubic structure of $\text{SrFeO}_{3-\delta}$ system. The unit cell parameter increases monotonically with increasing indium content in the samples, which is attributed both to the size effect from substituting $\text{Fe}^{3+/4+}$ ions with larger In^{3+} ions and to the accompanying decrease in the average oxidation state of iron.

Oxygen nonstoichiometry was determined using iodometric titration. Doping with In^{3+} leads to an increase in the concentration of oxygen vacancies at room temperature. Thermogravimetric measurements revealed that the release of oxygen from the crystal lattice begins at 300 °C, with samples containing high indium content exhibiting a lower oxygen loss upon heating. Thermal expansion was measured using high-temperature dilatometry. The temperature dependences of relative elongation are nonlinear.

Electrical conductivity measurements were carried out using the four-probe DC method in a controlled atmosphere ($p\text{O}_2 = 0.5\text{--}10^{-20}$ atm) in the temperature range 750–950 °C. The samples are characterized by mixed ionic-electronic conductivity throughout the entire temperature range studied. The maximum ionic conductivity value is observed for the composition $x = 0.05$. Upon further increase in indium content, the ionic conductivity decreases, which may be attributed to ordering of the crystal structure and partial blocking of the migration pathways for O^{2-} ions.

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